Updated October 12, 2026

# Jaehyun Park

Email | Website | LinkedIn

## Research Interests

Reality is abstract and high-dimensional. However, what we perceive are but fragments of it, colored by our own minds. From these fragments, how might we come to know the world more deeply?

*For a fuller account of my perspective, see my academic journey.*

## Education

**Seoul National University** Mar 2020 – expected Aug 2027

B.Sc. in Mathematics GPA: 4.02/4.3
B.Sc. in Mathematical Modeling for Complex Systems GPA: 4.3/4.3
B.Eng. in Architectural Engineering GPA: 3.96/4.3

Completed mandatory military service.

**Purdue University** Aug 2025 – Dec 2025

Exchange student in Mathematics and Industrial Engineering. GPA: 4.25/4.3

## Manuscripts in Preparation

**AAAAAAA (TBD)** Sep 2026 – Dec 2026 (Expected)

*In preparation with Dabeen Lee for submission to Mathematical Programming.*

**Inverse Variational Inequalities: Identification and Statistical Estimation** Feb 2026 – Nov 2026 (Expected)

*In preparation with Taewoo Lee for submission to Operations Research (INFORMS).*

This paper develops statistical estimators for equilibrium models from noisy observations, with consistency and asymptotic guarantees under different identification conditions.

## Research Experience

**Optimization and Algorithms Lab, SNU Math** Sep 2026 – present

*Advisor: Dabeen Lee*

- Studying looped Transformers and fixed-point reasoners.
- Co-authoring a manuscript.

**Operations Management, SNU Business**

*Advisor: Taewoo Lee*

*Research collaboration* Feb 2026 – present

Co-authoring a manuscript on identification and statistical estimation in inverse variational inequalities.

*Reading and study* Apr 2025 – Aug 2025

- Studied linear programming (Bertsimas) and convex optimization (Boyd).
- Read classic optimization papers from *Operations Research* and *Mathematical Programming*.

## Awards and Scholarships

**Dean's List and Semester Honors, Purdue University** Fall 2025

**Global Leadership Program Exchange Scholarship** Fall 2025

SNU College of Engineering; support for study abroad.

**SNU DAAE Alumni Scholarship for Academic Excellence** Fall 2023

Sole recipient, Department of Architecture and Architectural Engineering.

## Selected Projects

**Perturbation Analysis of a Pre-trained SPO Framework** Nov 2025 – Dec 2025

Investigating graph-model pre-training with conventional losses followed by SPO+ fine-tuning, and studying perturbation robustness through distributionally robust optimization and assumptions on graph topology.

**Operating Cost Optimization for Logistics Centers** 2024 Architecture Graduation Exhibition

Simulated demand in Seoul; used Gurobi and OSMnx to solve vehicle-routing and traveling-salesperson problems and visualize routes. Optimized photovoltaic installation area to reduce building energy costs.

## Service and Leadership

**Society of Engineering Network and Service** Sep 2023 – Feb 2025

Led engineering outreach for high-school students and STEM activities for middle-school students; participated in educational programs for underrepresented students.

**Student Council** Mar 2021 – Dec 2021

Managed digital communications and designed event announcements, campus banners, and graduation-exhibition posters.

## Skills

**Programming and tools:** Python, MATLAB, R, Julia, C/C++, LaTeX, Gurobi, OSMnx.

**Engineering software:** Midas, ETABS, Rhino, Grasshopper, AutoCAD.

**Languages:** Korean (native), English (advanced).

## Selected Coursework

**Seoul National University**

| Course | Term | Grade |
|---|---|---|
| Mathematical Algorithms 2 *(Graduate; Transformers)* | Fall 2026 | In progress |
| Topics in Mathematics 2 *(Analytic Number Theory)* | Fall 2026 | In progress |
| Algorithm | Summer 2026 | A+ |
| Measure Theory and Probability | Spring 2026 | A− |
| Introduction to Mathematical Analysis with Practice 1 | Spring 2026 | A+ |
| Differential Equations | Summer 2025 | A+ |
| Advanced Calculus 1 | Spring 2025 | A+ |
| Linear Algebra | Spring 2025 | A+ |
| AI & Big Data Basics | Spring 2025 | A0 |
| Microeconomics | Summer 2023 | A0 |
| Macroeconomics | Summer 2021 | A+ |

**Purdue University — 2025 Fall** Exchange

| Course | Code | Grade |
|---|---|---|
| Introduction to Convex Optimization *(Graduate)* | IE56100 | A+ |
| Linear Programming & Optimization Techniques | MA42100 | A0 |
| Operations Research – Stochastic Models | IE33600 | A+ |
| Probability | MA41600 | A+ |
| Introduction to Discrete Mathematics | MA37500 | A+ |
| Engineering Economics | IE34300 | A+ |